Shen Wang

M.Sc. Student in Mechanical Engineering (Mechatronics and Robotics Track)
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Research Interests

Robot learning and mechatronics for physical interaction, in particular physics-informed learning for contact-rich manipulation, with experiments on real robotic hardware.

Education

Politecnico di Milano 2024 – Jul 2027 (expected)
M.Sc. in Mechanical Engineering (Mechatronics and Robotics Track) *Milan, Italy*

- Selected courses (out of 30): Lab – Mechanical Engineering Applications of Deep Learning (30 cum laude), Micro and Nano Robotics (30 cum laude), Vision-Based 3D Measurements (30).

Dalian Maritime University 2020 – 2024
B.Eng. in Marine Engineering, average score 90.40/100 (GPA 4.04/5.0) *Dalian, China*

- Selected courses (out of 100): Fundamentals of Machinery (98), CAD/CAM/CAPP (97), Mechanics of Materials (97), Theoretical Mechanics (96), Intelligent Robot Systems (95), Mechanical Design (92).

Research and Project Experience

Physics-Informed Transformer for Contact-Force Prediction on a Franka Panda Feb – Jun 2026
Lab – Mechanical Engineering Applications of Deep Learning (Prof. Loris Roveda), Politecnico di Milano
Team project with Gabriele Lupo Pasini and Jiangquan Wei

- Extended a course-provided encoder–decoder Transformer with physics-informed components for three-axis contact-force prediction 0.24 s ahead.
- Collected motion and force data on an impedance-controlled Franka Panda under sponge and rigid-table contact; evaluated models using recordings from PoliMi and IDSIA.
- Used recent motion and measured-force history together with planned trajectories as model inputs; added a physics head for contact-parameter estimation, a physics-consistency loss, and a contact-model force embedding for the decoder.
- Compared spring–damper–mass, coupled-axis and Hunt–Crossley models against a data-driven baseline. Evaluated generalization from sponge to rigid-table contact and transfer from PoliMi to IDSIA recordings.

Planning, SLAM-Based Navigation and Intersection Management Oct 2025 – Feb 2026
Autonomous Vehicles (Prof. Stefano Arrigoni), Politecnico di Milano *Individual projects*

- Implemented RRT, RRT-Connect, RRT* and Informed RRT* in MATLAB and benchmarked them against A* on four maps at two grid resolutions (path length, planning time, memory; 50 seeded runs each).
- Built mapping (slam_toolbox), route planning and waypoint feedback control for a simulated TurtleBot3 (ROS 2 Jazzy, Gazebo, MATLAB); after the shortest route caused a tip-over at a tight corner, switched to clearance-aware route costs; added stop-and-replan with a heap-based A* for newly detected obstacles.
- Implemented a central manager that coordinates up to four vehicles at an unsignalized intersection, with right-of-way rules and deadlock detection and recovery; a first-come-first-served policy removed the observed deadlocks with only small changes in throughput and waiting time.

Synergistic Anti-Scaling with Triboelectric Nanogenerators and Copper Alloy Feb – Jun 2024
B.Eng. thesis (Prof. Guobin Li), Marine Engineering College, Dalian Maritime University

- Evaluated copper-alloy anti-scaling elements for ship cooling water that also act as solid–liquid triboelectric nanogenerators through a press-fitted PTFE block.
- Fabricated 60 copper-alloy test elements by wire EDM and built a 5 L stirred-tank rig with simulated hard cooling water; designed in SolidWorks and 3D-printed the impeller and an interference-fit motor bracket that removed the vibration seen in pre-tests.
- Ran five controlled 20-hour experiments (blank; alloy only; 10, 20 and 30 PTFE-fitted elements) with hourly coupon weighing and imaging; scale inhibition rose with the number of PTFE-fitted elements.

- This thesis built on a student project I led (Sep 2022 – Jun 2023); I am a co-inventor of the related Chinese invention patent CN116062913B (granted 2024).

Modular Underwater Vehicle (ROV/AUV) Oct 2021 – Sep 2023

Student team project, Dalian Maritime University

- Modelled, 3D-printed and assembled mechanical components for a modular ROV/AUV, iterating the prototypes within a student team.
- Funded by the National College Students' Innovation and Entrepreneurship Training Program (CNY 10,000).

Honors and Awards

2023 Outstanding Participant, Robotics and Intelligent Manufacturing Summer Camp, Shenyang Institute of Automation, Chinese Academy of Sciences
2023 Honorable Mention, Mathematical Contest in Modeling (MCM), COMAP
2022 Second Prize (Liaoning division), China Undergraduate Mathematical Contest in Modeling (CUMCM)
2022 First Prize, Liaoning Provincial Mathematical Modeling Contest
2022 Third Prize (national, individual), "Higher Education Cup" National College Students' Advanced Graphics Technology and Product Information Modeling Innovation Competition
2022 First Prize, Liaoning Provincial Mechanical Design Innovation Competition
2022 Third Prize, Northeast Three Provinces Mathematical Modeling Contest
2022 Second Prize (Liaoning division), "Datang Cup" National College Students' Mobile Communication 5G Technology Competition
2022 Honorable Mention, Interdisciplinary Contest in Modeling (ICM), COMAP

Skills and Languages

Programming & ML	Python, PyTorch, Weights & Biases, MATLAB, C (STM32, C51)
Robotics & simulation	ROS 2 (Jazzy), Gazebo, MuJoCo, Linux (Ubuntu 24.04)
Design & fabrication	SolidWorks, PTC Creo, AutoCAD, Ansys, Altair Inspire; 3D printing, wire EDM
Languages	Chinese (native); English (medium of instruction for my M.Sc.)
Certificates	Certified SOLIDWORKS Professional in Mechanical Design (CSWP), 2022

Extracurricular Activities

Student Leader, Innovation and Entrepreneurship Alliance Oct 2021 – Sep 2022

Dalian Maritime University

- Organised 12 innovation-and-entrepreneurship training sessions for first-year students, covering SolidWorks modelling, CAD drafting and MATLAB basics.

Volunteer, Youth Volunteer Association Dec 2021 – Oct 2023

Dalian Maritime University

- Community service (beach clean-ups, anti-fraud awareness sessions for senior citizens) and peer tutoring; Outstanding Volunteer at the 36th Liaoning Provincial Youth Science and Technology Innovation Competition.

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